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Automated 3D Stenosis Quantification and Visualization Workflow for Coronary Artery Disease Clinical Decision Support

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A la meva família i als amics que m'han acompanyat durant tot el grau, per la paciència, el suport i l'ànim en els moments més exigents.

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Agraïments

Primer de tot, vull agrair profundament al Pr. Oscar Camara Rey, director del treball a la Universitat Pompeu Fabra, per la seva orientació acadèmica, la confiança dipositada en mi i les revisions que han ajudat a donar forma i rigor a aquest projecte.

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Abstract

Coronary artery disease (CAD) assessment from coronary computed tomography angiography (CCTA) remains largely manual and time-consuming in clinical practice. At Hospital de la Santa Creu i Sant Pau, expert review of a single study can require between 30 minutes and two hours depending on case complexity. This bachelor's thesis presents an automated, end-to-end computational pipeline that transforms segmented coronary anatomy into quantitative stenosis metrics, CAD-RADS 2.0-oriented reporting support, and an interactive clinical visualization dashboard. The system is organized into four modular blocks: automated anatomy extraction, geometric stenosis quantification, segment labeling with CAD-RADS scoring, and a Streamlit-based decision-support interface. Designed as a transparent in-house framework rather than a proprietary black box, the pipeline aims to reduce reporting burden, improve reproducibility, and assist patient prioritization while preserving final clinical authority. Validation combines ASOCA and MACS-18 cohorts with synthetic phantoms for controlled geometric verification.

Resum

L'avaluació de la malaltia coronària (MAC) a partir de la tomografia computada coronària (TCCC) continua sent, en la pràctica clínica, un procés manual i molt costós en temps. A l'Hospital de la Santa Creu i Sant Pau, la revisió experta d'un estudi pot requerir entre 30 minuts i dues hores segons la complexitat del cas. Aquest treball de fi de grau presenta un pipeline computacional automatitzat d'extrem a extrem que transforma l'anatomia coronària segmentada en mètriques quantitatives d'estenosi, suport orientat al CAD-RADS 2.0 i un tauler clínic interactiu de visualització. El sistema s'organitza en quatre blocs modulares: extracció automatitzada de l'anatomia, quantificació geomètrica de l'estenosi, etiquetatge segmentari amb puntuació CAD-RADS i una interfície Streamlit de suport a la decisió. Dissenyat com a marc transparent i intern, el pipeline pretén reduir la càrrega de informes, millorar la reproduïbilitat i assistir la priorització de pacients sense substituir l'autoritat clínica final. La validació combina les cohorts ASOCA i MACS-18 amb fantasmes sintètics per a la verificació geomètrica controlada.

Resumen

La evaluación de la enfermedad coronaria (EC) a partir de la angiografía por tomografía computarizada coronaria (ATCC) sigue siendo, en la práctica clínica, un proceso manual y muy costoso en tiempo. En el Hospital de la Santa Creu i Sant Pau, la revisión experta de un estudio puede requerir entre 30 minutos y dos horas según la complejidad del caso. Este trabajo de fin de grado presenta un *pipeline* computacional automatizado de extremo a extremo que transforma la anatomía coronaria segmentada en métricas cuantitativas de estenosis, soporte orientado al CAD-RADS 2.0 y un panel clínico interactivo de visualización. El sistema se organiza en cuatro bloques modulares: extracción automatizada de la anatomía, cuantificación geométrica de la estenosis, etiquetado segmentario con puntuación CAD-RADS e interfaz Streamlit de apoyo a la decisión. Diseñado como un marco transparente e interno, el *pipeline* pretende reducir la carga de informes, mejorar la reproducibilidad y asistir la priorización de pacientes sin sustituir la autoridad clínica final. La validación combina las cohortes ASOCA y MACS-18 con fantomas sintéticos para la verificación geométrica controlada.

Prefaci

Aquest document recull el treball de fi de grau realitzat durant el curs 2025–2026 en el marc del Grau en Enginyeria Matemàtica en Ciència de Dades de la Universitat Pompeu Fabra, en col·laboració amb el Dimension Lab de l'Hospital de la Santa Creu i Sant Pau.

El cos principal del treball (Introducció a Conclusions) està redactat en anglès per coherència amb la literatura científica del domini. Les seccions preliminars obligatòries (resums i agraïments) es presenten en català, anglès i castellà segons els requisits de la plantilla UPF.

Declaració sobre l'ús de la intel·ligència artificial. D'acord amb la instrucció universitària sobre l'ús de la IA en activitats avaluables, declaro que el contingut tècnic i clínic del treball — especialment la Introducció, la Metodologia i les Conclusions — ha estat redactat i revisat per mi, amb el suport del meu director i supervisor. Per a la integració en LaTeX, la correcció de format i l'esborrany de les seccions preliminars (agraïments, resums i prefaci), he utilitzat eines d'IA generativa com a assistent d'edició i maquetació. Tots els fragments generats o reformulats amb IA han estat revisats, validats i adaptats per mi abans de la seva inclusió. Qualsevol text, figura o referència inclosa sense revisió pròpia serà clarament identificada en versions futures del document si així ho requereix el director del treball.

Nota sobre figures i dades. Les figures de fluxos clínics adaptades de treballs interns de l'hospital es citen explícitament al peu de figura. Les dades de cohorts clíniques i els resultats del pipeline es documentaran als capítols corresponents i, quan calgui, a l'apèndix de suport.

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Chapter 1

INTRODUCTION

1.1. Clinical Context and Motivation

1.1.1. Anatomy of Coronary Arteries

The human heart is the central organ of the cardiovascular system and requires a continuous supply of oxygen and nutrients to function properly. This supply is provided by the coronary artery tree, a specialized vascular network that delivers blood directly to the heart muscle, known as the myocardium. The coronary arteries originate from the root of the aorta and surround the surface of the heart in a crown-like arrangement, from which the term *coronary artery* is derived [Cleveland Clinic, 2024]. As illustrated in Figure 1.1, the coronary arterial system is mainly composed of two vessels: the right coronary artery (RCA) and the left coronary artery (LCA).

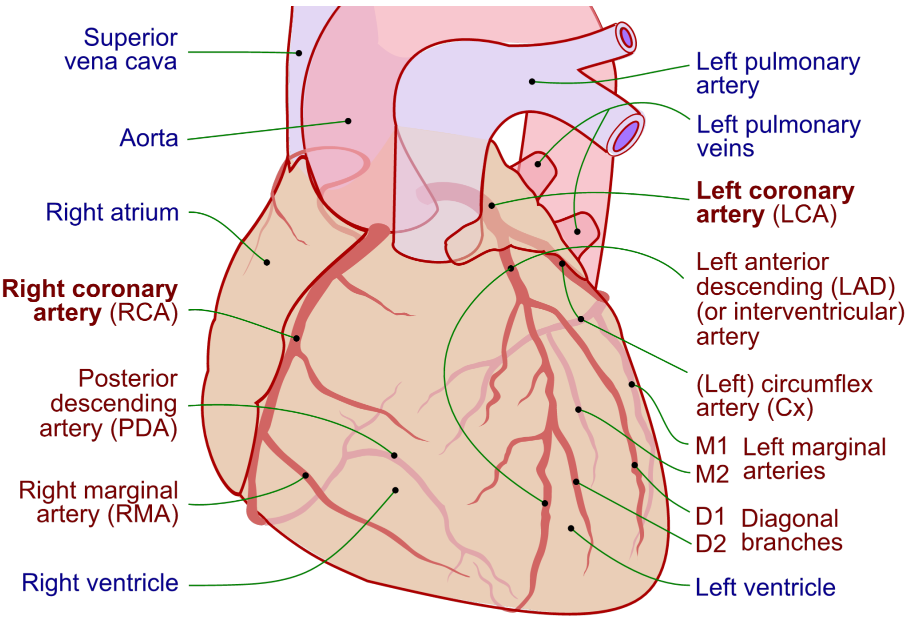

Figure 1.1: Anterior view of the human coronary tree. Coronary arteries are highlighted in red, and other cardiac structures in blue. Adapted from Lynch et al. [Lynch et al., 2010].

The RCA primarily supplies blood to the right atrium and right ventricle. In contrast, the LCA quickly divides into two major branches: the left anterior descending (LAD) artery and the circumflex (LCx), which together supply the majority of the left ventricle and the interventricular septum [Cleveland Clinic, 2024]. From these primary vessels, a network of smaller branches extends deep into the myocardium to ensure regional blood delivery across the entire heart.

A critical characteristic of the coronary circulation is its limited collateral connectivity. Unlike other vascular systems in the human body, coronary arteries generally lack sufficient alternative pathways to redirect blood flow when an obstruction occurs. Consequently, any narrowing or blockage in a proximal arterial segment directly compromises blood flow to all downstream regions supplied by that vessel [Seiler, 2010].

Additionally, coronary anatomy presents significant inter-patient variability in terms of vessel size, branching structure, and coronary dominance. Dominance is defined by which artery gives rise to the posterior descending artery (PDA) and can be classified as right-dominant, left-dominant, or codominant. Right-dominant circulation, in which the PDA originates from the RCA, is the most common configuration, followed by left dominance and codominance [Wu et al., 2024]. Understanding this complex anatomical layout is fundamental, as it provides the basis for locating lesions, quantifying stenosis severity, and stratifying risk in coronary artery disease (CAD) diagnosis.

1.1.2. Coronary Artery Disease

Coronary artery disease (CAD) remains one of the leading causes of morbidity and mortality worldwide [World Health Organization, 2023]. It is characterized by the progressive accumulation of atherosclerotic plaque within the walls of the coronary arteries, leading to a narrowing of the vessel lumen known as stenosis. This narrowing restricts blood flow to the myocardium and, if left untreated, can result in myocardial ischemia or infarction [Knuuti et al., 2020, McCullough, 2007]. Figure 1.2 illustrates this pathological mechanism, contrasting a normal artery with one obstructed by plaque buildup.

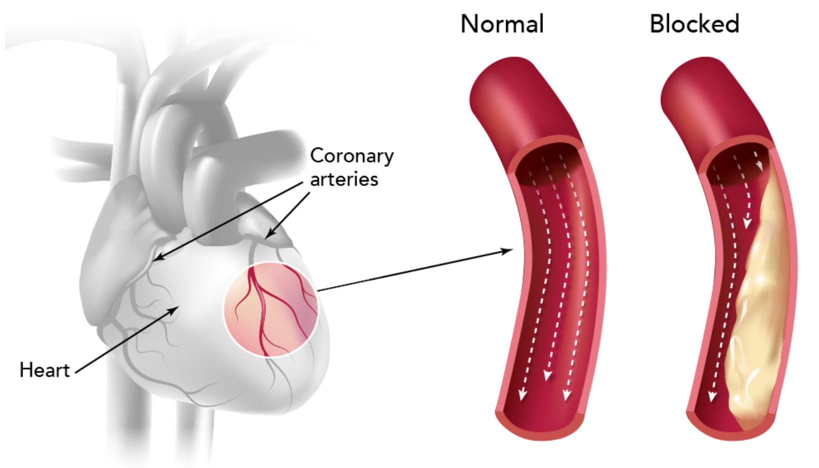

Figure 1.2: Coronary arteries narrowed by atherosclerotic plaque. Adapted from Kaiser Permanente [Kaiser Permanente, 2024].

Early and accurate diagnosis is essential for effective risk stratification and patient management. In current clinical practice, coronary computed tomography angiography (CCTA) is the standard non-invasive imaging technique for CAD evaluation [Knuuti et al., 2020]. CCTA provides detailed 3D reconstructions of the coronary tree, allowing clinicians to detect plaques, analyze their spatial location, and estimate the degree of luminal narrowing.

However, despite the high resolution of CCTA, the diagnostic workflow remains highly time-consuming and heavily dependent on manual and visual assessment. For instance, at Hospital de la Santa Creu i Sant Pau, reporting non-pathological or low-complexity cases requires approximately 30 minutes, while complex evaluations can take up to 2 hours [Acebes Pinilla, 2024]. This significant clinical burden drives the need for automated, data-driven tools capable of quantifying stenosis efficiently and supporting clinical decision-making, while fully preserving expert interpretability and clinical control.

1.1.3. Stenosis Quantification: A Key Indicator of CAD Severity

Among the various indicators used to assess coronary artery disease, stenosis severity plays a central role in clinical diagnosis and risk stratification. Stenosis represents the reduction of the vessel lumen caused by atherosclerotic plaque accumulation, a condition directly associated with an increased risk of myocardial ischemia and adverse cardiac events [Knuuti et al., 2020, Garcia-Garcia et al., 2014]. In clinical practice, this severity is quantified using geometric measurements such as the minimal lumen area (MLA) or percentage diameter stenosis (%DS). These metrics evaluate the degree of narrowing by comparing the most constricted point of a lesion with a reference baseline, which is typically estimated from adjacent, relatively healthy vessel segments [Garcia-Garcia et al., 2014, Hideo-Kajita et al., 2019].

Although these definitions are conceptually simple, accurate stenosis quantification presents significant clinical challenges. Coronary arteries exhibit complex, tortuous geometries, and diffuse atherosclerosis often makes it difficult to identify truly healthy reference regions. In addition, limitations in image resolution, segmentation errors, and noise in CCTA-derived data can highly influence measurements, leading to variability in stenosis estimates [Kalisz et al., 2016, Hideo-Kajita et al., 2019].

In routine clinical practice, stenosis assessment is often performed visually or using semi-automatic tools provided by proprietary software [Acebes Pinilla, 2024, Cury et al., 2022]. Although these methods are widely used, they depend heavily on the clinician's interpretation and can vary between observers, which reduces consistency and reproducibility [Hideo-Kajita et al., 2019].

Consequently, there is a strong clinical need to develop robust, transparent, and reproducible quantification methods. Automating these geometric measurements has the potential to standardize stenosis evaluation, reduce reporting time, and provide reliable quantitative data essential for downstream clinical tasks, such as CAD-RADS 2.0 assignment and patient prioritization.

1.1.4. CAD-RADS 2.0 and the Need for Decision Support

To standardize the reporting of coronary disease and facilitate clinical communication, the Coronary Artery Disease-Reporting and Data System (CAD-RADS 2.0) was established [Cury et al., 2022]. This structured framework categorizes patients into distinct risk scores ranging from 0 to 5, primarily based on the most severe stenosis measured via CCTA. Each categorical score reflects the overall disease burden and provides specific clinical recommendations, offering a standardized pathway for subsequent patient management [Cury et al., 2022].

While CAD-RADS 2.0 significantly improves reporting consistency across institutions, its assignment still relies heavily on the manual interpretation of stenosis severity across multiple vessels and segments. This process requires clinicians to continuously synthesize quantitative geometric measurements with complex anatomical context. Consequently, evaluating these scans becomes highly cognitively demanding, particularly in time-constrained clinical environments [Acebes Pinilla, 2024]. In practice, manual scoring remains susceptible to inter-observer variability, especially in borderline cases or in patients presenting with multiple moderate lesions. Such discrepancies can lead to inconsistent diagnostic classifications, ultimately impacting downstream clinical decisions, further testing, and patient prioritization [Acebes Pinilla, 2024, Hideo-Kajita et al., 2019].

To address these limitations, there is a growing clinical imperative to develop automated decision support tools. These solutions can span from rule-based algorithms mapping quantitative stenosis measurements to machine learning approaches that leverage structured imaging features. Crucially, these computational tools aim to provide objective, reproducible, and transparent suggestions to assist radiologists and cardiologists. By streamlining the CAD-RADS assignment process, these systems reduce reporting times and variability while strictly preserving expert clinical judgment and final decision authority.

1.2. Clinical Automation Challenges at Hospital de la Santa Creu i Sant Pau

1.2.1. Current CAD Diagnostic Workflow and Associated Limitations

The current diagnostic workflow for Coronary Artery Disease (CAD) at Hospital de la Santa Creu i Sant Pau involves a complex and coordinated series of steps, managed primarily by the specialized cardiac imaging unit (Unitat d'Imatge i Funció Cardíaca, UIFC). As illustrated in Figure 1.3, the clinical pathway begins when a patient presenting with symptoms such as chest pain is scheduled for a Coronary Computed Tomography Angiography (CCTA). Once the scan is acquired, the images are stored in the hospital's *Picture Archiving and Communication System* (PACS) for further evaluation.

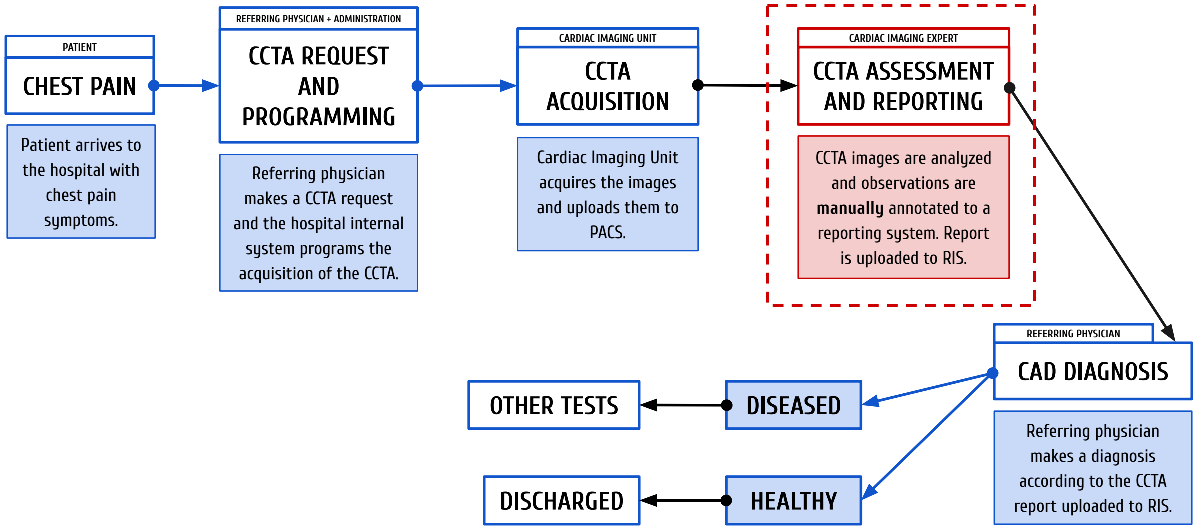

Figure 1.3: Current CAD diagnostic workflow at Hospital de Sant Pau. The primary operational bottleneck (image assessment and reporting) is highlighted in red. Adapted from Ferrer Beltran [Ferrer Beltran, 2024].

The core of this diagnostic pipeline relies heavily on the visual and manual analysis of these scans by expert cardiologists and radiologists to extract clinical metrics. These findings are then compiled into a clinical report, which the referring clinician evaluates to determine the appropriate subsequent medical actions, ranging from discharging a healthy patient to programming further invasive tests.

While the overall infrastructure is well-established, the central phase of image analysis and reporting (highlighted in red in Figure 1.3) represents a critical operational bottleneck.

1.2.2. The Bottleneck: Manual Image Analysis and Reporting

Zooming into the specific image analysis and reporting phase (Figure 1.3), the limitations of the current clinical pathway become evident. Currently, clinicians rely on commercial software solutions, such as *syngo.via*, to manually inspect the coronary arteries, differentiate segments, and calculate stenosis severity. Although these platforms offer semi-automated tools, the process remains highly dependent on the clinician's expertise to visually verify the vessels, identify bifurcations, and assess the plaque burden. Depending on the anatomical complexity and image quality, analyzing a single patient can consume around 30 minutes for healthy cases and 90 minutes for complex cases [Acebes Pinilla, 2024].

Once the visual assessment is complete, the extracted metrics are manually entered into an internal custom-built platform called *Filemaker*, which generates a template-based preliminary report. The specialist must then review, correct, and finalize this text before it is integrated into the *Radiology Information System* (RIS).

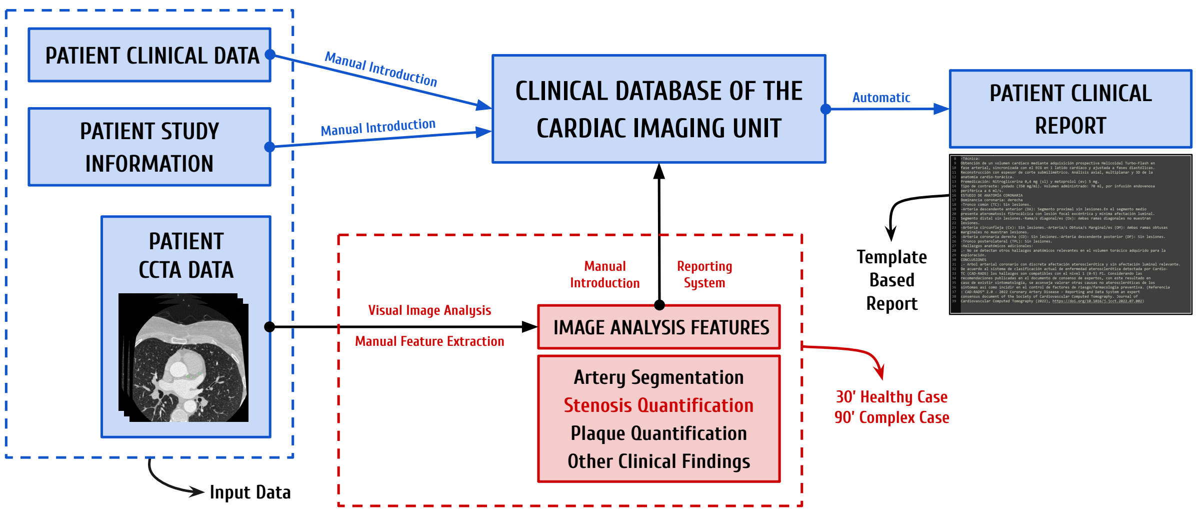

Figure 1.4: Detailed representation of the image analysis and reporting workflow at Hospital de Sant Pau. Adapted from Acebes Pinilla [Acebes Pinilla, 2024].

This heavy reliance on manual quantification and data entry presents significant operational challenges. Consequently, highly specialized experts spend considerable time performing repetitive, manual tasks on low-risk patients, rather than focusing their expertise on critical cases. This lack of an automated, prioritized workflow delays the diagnosis of severe cases and highlights a clear necessity for robust and automated solutions.

1.3. State of the Art: Automation of CAD Diagnosis and Visualization

The integration of Artificial Intelligence (AI) and advanced computer vision in cardiac imaging has seen exponential growth over the past decade, driven by the need to optimize clinical workflows [Litjens et al., 2017]. Deep learning algorithms, particularly Convolutional Neural Networks (CNNs), have demonstrated remarkable success in medical image analysis, encouraging a move from manual CCTA assessment to automated systems [Slomka et al., 2017]. However, significant limitations remain regarding their practical implementation. For these new techniques to succeed, they must be practical for hospitals, integrate smoothly into current systems, and be easily adopted by the clinical staff [Kelly et al., 2019].

The prerequisite first step in any automated CAD assessment pipeline is the accurate segmentation and anatomical labeling of the coronary artery tree; before any downstream diagnostic metric can be calculated, the vessels must be correctly isolated and identified. Recent advancements have leveraged deep learning architectures, such as 3D U-Nets, to extract the vessel lumen from CCTA images [Zhang et al., 2024, Çiçek et al., 2016], while graph-based models like Graph Convolutional Networks (GCNs) are increasingly used to ensure topological consistency when labeling segments according to standard clinical guidelines [Yang et al., 2020]. Within Hospital de la Santa Creu i Sant Pau, preliminary research has explored these initial segmentation and labeling tasks using transformer-based models and the nn-UNet framework [Clapers Colet, 2025, Sanchez Gomez, 2022].

Following successful segmentation, extracting the precise geometric properties of the vessels is essential. The Vascular Modeling Toolkit (VMTK) stands out as one of the most prominent open-source libraries for medical geometry data extraction [Antiga et al., 2008]. Specifically, algorithms implemented within VMTK are widely utilized to extract accurate vascular centerlines and compute cross-sectional areas. These metrics provide the essential topological framework required to perform continuous Multi-Planar Reformations and subsequently evaluate vessel narrowing.

Once the 3D geometry of the coronary tree is established, the focus shifts to quantifying the disease burden. Current stenosis quantification techniques range from traditional intensity-based thresholding (which often struggles with severe calcifications) to modern machine learning models that predict stenosis severity directly from image features [Zhang et al., 2024, Hong et al., 2019]. Alternatively, geometric, non-ML approaches offer highly interpretable and robust methods by calculating the percentage of diameter or area stenosis based purely on the physical vessel profile, comparing the minimal lumen area against an interpolated healthy reference [Hideo-Kajita et al., 2019]. Recognizing the clinical value of algorithmic transparency, ongoing research at Hospital de la Santa Creu i Sant Pau has actively explored these purely geometric techniques to quantify stenosis, establishing a reliable, explainable foundation for disease assessment [Burrull, 2023].

Beyond quantification, the effective visualization of these complex 3D metrics remains a critical challenge in the state of the art [Borkin et al., 2011]. In the current clinical routine at Hospital de Sant Pau, specialists must manually navigate through cross-sectional slices and use standard viewing software to visually synthesize the data, assess the lesions, and manually fill out structured reports [Acebes Pinilla, 2024]. This cognitive load highlights a significant gap: the lack of dedicated, interactive visualization tools that bridge the gap between automated metrics and clinical reality. Modern research and clinical consensus emphasize that integrating 3D anatomical models alongside continuous luminal profiles and real medical images (such as Curved Multi-Planar Reformations) into a unified graphical interface is essential. Such integration allows clinicians to continuously validate automated findings against the ground-truth CCTA, drastically reducing the time and effort required to verify stenosis locations and seamlessly generate diagnostic reports [Borkin et al., 2011].

Several commercial software solutions have emerged to address these overarching needs, including platforms like *Cleerly* [Cleerly, Inc., 2024] and *Artrya Salix* [Artrya Ltd., 2024], which use AI to analyze arterial plaque and assess ischemia risk from CCTA images. While these robust dashboards offer highly desirable visualization components, they often function as proprietary "black boxes". This lack of algorithmic transparency significantly hinders explainability, a feature strongly demanded by medical experts to verify how a specific CAD-RADS score, plaque distribution, or stenosis percentage was computed. Finally, these standalone commercial platforms frequently struggle to integrate well into the highly specific IT infrastructures and patient prioritization workflows of public hospitals [Kelly et al., 2019].

1.4. Project Scope and Contributions

To address the clinical challenges outlined in Section 1.2, and to overcome the limitations of commercial tools discussed in Section 1.3, this project contributes directly to the comprehensive, transparent, and in-house AI-enabled diagnostic framework developed by the Dimension Lab at Hospital de la Santa Creu i Sant Pau [Acebes Pinilla, 2024]. While previous academic efforts have tackled earlier stages of this pipeline, such as patient prioritization [Ferrer Beltran, 2024] and coronary artery segmentation and labeling [Clapers Colet, 2025], a critical gap remains in translating raw segmented geometries into actionable clinical insights.

Therefore, the scope of this bachelor's thesis focuses on automating the geometric quantification of coronary stenosis, predicting the associated CAD-RADS score, and integrating these findings into an interactive clinical dashboard.

Figure 1.5 provides a high-level overview of where this project sits within the pipeline, while the detailed technical workflow is reserved for the Methodology chapter.

By seamlessly connecting raw image analysis with the hospital's reporting systems, this project aims to mitigate operational bottlenecks and reduce overall diagnostic times.

However, it is crucial to emphasize that the solution developed in this project is intended solely as a clinical decision-support tool, not as a replacement for medical expertise. In the highly sensitive healthcare sector, human oversight is crucial. Rather than replacing detailed diagnostic software, this visualization tool is designed to act as a rapid initial assessment interface. By presenting an immediate, high-level overview of the patient's critical metrics, it assists specialists in quickly evaluating prioritized patient lists, helping them decide which cases require immediate, in-depth analysis. Thus, the automated findings are designed to streamline the workflow, ensuring that the final diagnostic decisions and clinical

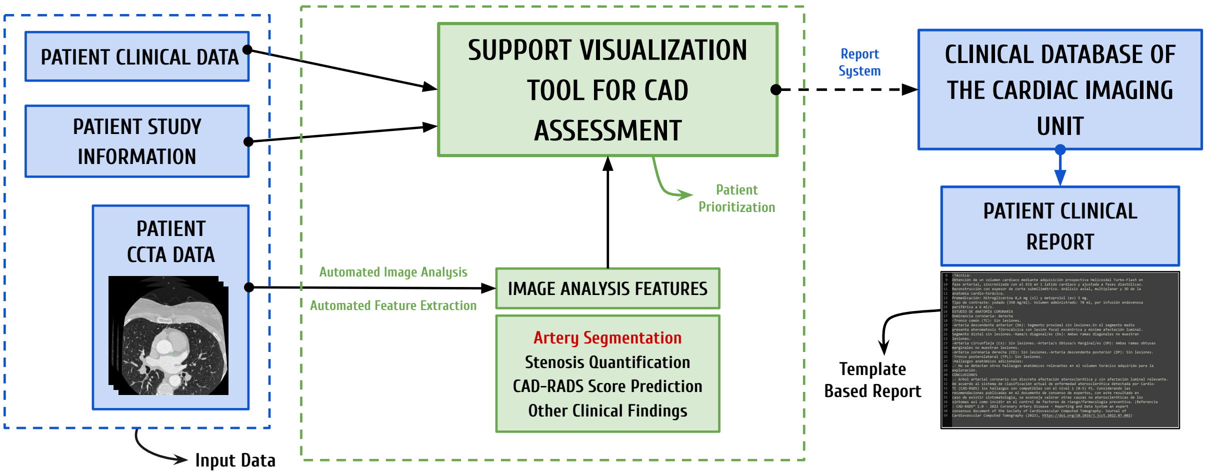

Figure 1.5: Contextual overview of the proposed automated diagnostic framework at Hospital de la Santa Creu i Sant Pau. The contributions of this thesis (green dashed box) bridge the gap between existing hospital data systems (blue) and the final clinical report. Adapted from Acebes Pinilla [Acebes Pinilla, 2024].

validations always remain firmly in the hands of the medical professionals.

1.5. Objectives

The primary objective of this thesis is to design and develop an automated, end-to-end pipeline that successfully bridges the current gap in the CAD diagnostic workflow. Importantly, this project does not seek to maximize the precision of individual algorithmic components; rather, the proposed functional workflow is, in itself, the core product of this work.

To achieve this overarching goal, the project addresses a sequence of intermediate objectives. First, it requires deep contextualization within the complex clinical environment to evaluate prior research conducted at the hospital and design a unified, clinically applicable framework. Building upon this foundation, the project aims to implement computational geometry algorithms to accurately extract cross-sectional areas and calculate the percentage of area stenosis from 3D models. Subsequently, a standardized logic is developed to translate these geometric metrics into a reliable CAD-RADS score prediction.

To aggregate these findings and provide cardiologists with interpretable metrics, an interactive clinical dashboard is built. Functioning as a rapid-access pop-up interface, this Support Visualization Tool gives clinicians an immediate, visual summary of the patient's coronary status, ultimately reducing medical workload, optimizing case triaging, and significantly decreasing overall assessment time. Finally, a technical validation is conducted to verify the reliability of the extracted metrics and ensure the overall stability of the complete framework.

Chapter 2

METHODOLOGY

Chapter 3

RESULTS

3.1. Pipeline Execution on ASOCA

3.2. Pipeline Execution on MACS-18

3.3. Geometric and Algorithmic Accuracy via Synthetic Data

3.4. Coronary Artery Dashboard Visualization Output

3.5. User Experience and Interface Usability Evaluation

Chapter 4

DISCUSSION

Chapter 5

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Appendix A

SUPPORTING MATERIAL